The Weil Explicit Formula in the Meixner–Pollaczek Basis

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Abstract

We evaluate the Weil explicit-formula quadratic form on the polarized Meixner–Pollaczek basis at $\lambda = \frac{1}{4}$, whose weight $|\Gamma(\frac{1}{4} + ix)|^2$ is the archimedean Euler factor of ζ on the critical line. Every matrix entry is exactly computable: the pole term is rational in $\sqrt{\pi}$, the archimedean term lies in the rank-two lattice $\mathbb{Q}\sqrt{2\pi} \oplus \mathbb{Q}\sqrt{2\pi}\,(\gamma + \log 8\pi)$, and the prime term is a rational series in the values $-\zeta'/\zeta(1+2k)$, $k \geq 1$, with no prime sum remaining. The Riemann hypothesis implies every truncation of the resulting explicit matrix is positive semidefinite.

1 The form

Let P_m be the Meixner–Pollaczek polynomials with $\lambda = \frac{1}{4}$, $\phi = \frac{\pi}{2}$:

$$2t\,P_m = (m+1)P_{m+1} + (m - \tfrac{1}{2})P_{m-1}, \qquad P_0 = 1, \quad P_1 = 2t,$$

orthogonal on $\mathbb{R}$ for the weight $W(t) = |\Gamma(\frac{1}{4} + it)|^2$ with norms $h_m = 2\pi\,\Gamma(m + \frac{1}{2})/(\sqrt{2}\,m!)$, and write $\widehat{P}_m = P_m/\sqrt{h_m}$. For $m \equiv n \pmod 2$ define the even test functions

$$h_{mn}(r) \;=\; \big(r^2 + \tfrac{1}{4}\big)\,\widehat{P}_m(\tfrac{r}{2})\,\widehat{P}_n(\tfrac{r}{2})\,\big|\Gamma(\tfrac{1}{4} + \tfrac{ir}{2})\big|^2, \qquad Q_{mn}(r) \;=\; \big(r^2 + \tfrac{1}{4}\big)\,P_m(\tfrac{r}{2})\,P_n(\tfrac{r}{2}) \;\in\; \mathbb{Q}[r^2].$$

The factor $r^2 + \frac{1}{4}$ cancels the poles of the weight at $r = \pm\frac{i}{2}$, so h_{mn} is analytic on $|\operatorname{Im} r| < \frac{5}{2}$ and decays like $e^{-\pi|r|/2}$: it is admissible for the explicit formula. With $g(u) = \frac{1}{2\pi}\int_{\mathbb{R}} h(r)e^{-iur}\,dr$, the formula for ζ reads (Iwaniec–Kowalski, Thm. 5.12)

$$\sum_{\rho} h(\gamma_\rho) = h(\tfrac{i}{2}) + h(-\tfrac{i}{2}) - g(0)\log\pi + \frac{1}{2\pi}\int_{\mathbb{R}} h(r)\,\operatorname{Re}\psi\big(\tfrac{1}{4} + \tfrac{ir}{2}\big)dr - 2\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\,g(\log n),$$

the sum running over nontrivial zeros $\rho = \frac{1}{2} + i\gamma_\rho$. Set $W_{mn} = \sum_\rho h_{mn}(\gamma_\rho)$.

Theorem 1.1. *For $m \equiv n \pmod 2$,*

$$\sqrt{h_m h_n}\,W_{mn} \;=\; 2\sqrt{\pi}\,q_{mn} + \sqrt{2\pi}\,\Big[\alpha_{mn} + \beta_{mn}\big(\gamma + \log 8\pi\big)\Big] - 4\sqrt{\pi}\sum_{k\geq 1} a_k^{(mn)}\Big(-\frac{\zeta'}{\zeta}(1+2k)\Big),$$

where γ is Euler's constant and $q_{mn}, \alpha_{mn}, \beta_{mn}, a_k^{(mn)} \in \mathbb{Q}$ are explicitly computable (Lemmas 2.1–2.4); the series converges geometrically ($|a_k| = O(k^{m+n})$ against $-\zeta'/\zeta(1+2k) \sim \log 2 \cdot 4^{-k}$).

Corollary 1.2. *$\beta_{mn} = 0$ whenever $|m - n| \geq 4$: those archimedean entries are pure rational multiples of $\sqrt{2\pi}$.*

2 Proofs

Lemma 2.1 (pole term)**.** $\sqrt{h_m h_n}\,\big[h_{mn}(\frac{i}{2}) + h_{mn}(-\frac{i}{2})\big] = 2\sqrt{\pi}\,q_{mn}$ *with* $q_{mn} = P_m P_n(\frac{i}{4}) + P_m P_n(-\frac{i}{4}) \in \mathbb{Q}$.

Proof. Near $r = \frac{i}{2}$ put $r = \frac{i}{2} + \varepsilon$: the first Gamma factor is $\Gamma(\frac{i\varepsilon}{2}) \sim \frac{2}{i\varepsilon}$ while $r^2 + \frac{1}{4} \sim i\varepsilon$, so their product tends to 2; the second factor is $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Hence $h_{mn}(\frac{i}{2}) = 2\sqrt{\pi}\,\widehat{P}_m\widehat{P}_n(\frac{i}{4})$, and likewise at $-\frac{i}{2}$. Since P_m has rational coefficients, $P_m P_n(\frac{i}{4}) + P_m P_n(-\frac{i}{4})$ is rational. □

Lemma 2.2 (Fourier transform of the weight)**.** *For* $\lambda > 0$,

$$\frac{1}{2\pi}\int_{\mathbb{R}} \big|\Gamma(\lambda + \tfrac{ir}{2})\big|^2 e^{-iur}\,dr \;=\; 2\,\Gamma(2\lambda)\,(2\cosh u)^{-2\lambda} \;=\; \frac{\Gamma(\lambda)\Gamma(\lambda+\frac{1}{2})}{\sqrt{\pi}}\,\operatorname{sech}^{2\lambda} u.$$

Proof. Substituting $v = e^{-2u}$ in $\int_{\mathbb{R}} (2\cosh u)^{-2\lambda} e^{-iur}\,du$ gives $\frac{1}{2}\int_0^\infty v^{\lambda + ir/2 - 1}(1+v)^{-2\lambda} dv = \frac{1}{2}B(\lambda + \frac{ir}{2}, \lambda - \frac{ir}{2}) = |\Gamma(\lambda + \frac{ir}{2})|^2/(2\Gamma(2\lambda))$; Fourier inversion gives the first equality, and the Legendre duplication formula $\Gamma(2\lambda) = 4^\lambda \Gamma(\lambda)\Gamma(\lambda + \frac{1}{2})/(2\sqrt{\pi})$ the second. □

Two consequences. First, taking $2k$ derivatives at $u = 0$: with $\operatorname{sech}^{2\lambda} u = \sum_j \sigma_j(\lambda) u^{2j}$, $\sigma_j \in \mathbb{Q}[\lambda]$,

$$\widetilde{\mu}_k(\lambda) := \int_{\mathbb{R}} r^{2k} \big|\Gamma(\lambda + \tfrac{ir}{2})\big|^2 dr = 2\sqrt{\pi}\,(-1)^k (2k)!\,\sigma_k(\lambda)\,\Gamma(\lambda)\Gamma(\lambda+\tfrac{1}{2}) : \tag{1}$$

every moment is a $\mathbb{Q}[\lambda]$-multiple of $\Gamma(\lambda)\Gamma(\lambda+\frac{1}{2})$. Second, multiplication by r under the integral is $i\,d/du$ outside, and on functions $p(\tanh u)\operatorname{sech}^{1/2} u$ the derivative acts as $p \mapsto (1-t^2)p' - \frac{t}{2}\,p$ on $\mathbb{Q}[t]$.

Lemma 2.3 (archimedean term)**.** *Write* $Q_{mn}(r) = \sum_k q_k r^{2k}$. *Then*

$$\sqrt{h_m h_n}\Big[\frac{1}{2\pi}\int_{\mathbb{R}} h_{mn}(r)\,\operatorname{Re}\psi(\tfrac{1}{4} + \tfrac{ir}{2})\,dr - g_{mn}(0)\log\pi\Big] = \sqrt{2\pi}\,\big[\alpha_{mn} + \beta_{mn}(\gamma + \log 8\pi)\big]$$

with $\alpha_{mn} = \frac{1}{2}\sum_k q_k (-1)^k (2k)!\,\sigma_k'(\frac{1}{4})$ *and* $\beta_{mn} = -\sum_k q_k (-1)^k (2k)!\,\sigma_k(\frac{1}{4}) = -\frac{1}{2\sqrt{2}\,\pi^{3/2}}\int_{\mathbb{R}} Q_{mn}(r)\,\big|\Gamma(\frac{1}{4} + \frac{ir}{2})\big|^2 dr$.

Proof. Since $2\operatorname{Re}\psi(\lambda+ix)\,|\Gamma(\lambda+ix)|^2 = \partial_\lambda |\Gamma(\lambda+ix)|^2$, the digamma integral is $\frac{1}{2}\sum_k q_k\,\widetilde{\mu}_k'(\frac{1}{4})/(2\pi)$. Differentiate (1): $\widetilde{\mu}_k' = 2\sqrt{\pi}(-1)^k(2k)!\,\Gamma(\lambda)\Gamma(\lambda+\frac{1}{2})[\sigma_k' + \sigma_k(\psi(\lambda) + \psi(\lambda+\frac{1}{2}))]$. At $\lambda = \frac{1}{4}$: $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2}$ and $\psi(\frac{1}{4}) + \psi(\frac{3}{4}) = -2\gamma - 6\log 2$ (the reflection and duplication values $\psi(\frac{1}{4}) = -\gamma - 3\log 2 - \frac{\pi}{2}$, $\psi(\frac{3}{4}) = -\gamma - 3\log 2 + \frac{\pi}{2}$). The $\log\pi$ term contributes $-\log\pi\sum_k q_k \widetilde{\mu}_k(\frac{1}{4})/(2\pi)$. Collecting, with $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4})/\pi = \sqrt{2}$ turning the prefactor into $\sqrt{2\pi}$, the constants $\gamma + 3\log 2$ and $\log\pi$ appear with the same rational coefficient β_{mn}, giving the single constant $\gamma + 3\log 2 + \log\pi = \gamma + \log 8\pi$. □

Proof of the Corollary. β_{mn} is proportional to $\int (4t^2 + \frac{1}{4}) P_m P_n\,W\,dt$, and $\deg\,(4t^2 + \frac{1}{4})P_m = m + 2 < n$ when $n \geq m + 4$, so the integral vanishes by orthogonality. □

Lemma 2.4 (prime term)**.** *Let* $T_{mn} \in \mathbb{Q}[t]$ *be the image of* 1 *under* $Q_{mn}(i\,\mathcal{D})$, *where* $\mathcal{D}\colon p \mapsto (1-t^2)p' - \frac{t}{2}p$. *Then*

$$g_{mn}(u) = \frac{\sqrt{2\pi}}{\sqrt{h_m h_n}}\,T_{mn}(\tanh u)\,\operatorname{sech}^{1/2} u, \qquad T_{mn}(1) = 0,$$

and, with $(1+w)^{-1/2}\,T_{mn}\big(\frac{1-w}{1+w}\big) = \sum_{k\geq 1} a_k^{(mn)} w^k$,

$$2\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\,g_{mn}(\log n) = \frac{4\sqrt{\pi}}{\sqrt{h_m h_n}}\sum_{k\geq 1} a_k^{(mn)}\Big(-\frac{\zeta'}{\zeta}(1+2k)\Big).$$

Proof. The closed form for g_{mn} is Lemma 2.2 at $\lambda = \frac{1}{4}$ together with $r \leftrightarrow i\,d/du$. For $T_{mn}(1) = 0$: h_{mn} is analytic on $|\operatorname{Im} r| < \frac{5}{2}$ (the pole at $r = -\frac{i}{2}$ is cancelled), so shifting the contour gives $g_{mn}(u) = O(e^{-cu})$ for every $c < \frac{5}{2}$, while $\operatorname{sech}^{1/2} u \sim \sqrt{2}\, e^{-u/2}$; the coefficient of $e^{-u/2}$, namely $T_{mn}(1)$, must vanish. At $u = \log n$ one has $\tanh = \frac{1-w}{1+w}$ and $n^{-1/2} \operatorname{sech}^{1/2} = \sqrt{2}\, n^{-1}(1+w)^{-1/2}$ with $w = n^{-2}$, so the prime sum becomes $\sum_n \Lambda(n) n^{-1} \sum_k a_k n^{-2k}$; since $a_0 = 0$ (this is $T_{mn}(1) = 0$) the double sum converges absolutely and swapping gives the ζ'/ζ series. □

Theorem 1.1 is the explicit formula assembled from Lemmas 2.1, 2.3, 2.4. □

3 Numerical certification

All three sides were computed independently and compared with $2\sum_{\gamma_j \le 85} h_{mn}(\gamma_j)$ over the first 23 zeros of ζ (whose weight bounds the truncation by $e^{-85\pi/2} \approx 10^{-58}$). For all 16 entries with $m, n \le 6$ the identity of Theorem 1.1 holds to between 39 and 56 digits, and the lattice memberships of Lemma 2.3 were confirmed by PSLQ at 125 digits. Sample exact archimedean entries $u_{mn} = \sqrt{h_m h_n}\, A_{mn}$:

$$u_{00} = \sqrt{2\pi}\Big[1 - \tfrac{3}{4}(\gamma + \log 8\pi)\Big], \quad u_{11} = \sqrt{2\pi}\Big[\tfrac{21}{4} - \tfrac{15}{8}(\gamma + \log 8\pi)\Big], \quad u_{04} = \tfrac{3}{16}\sqrt{2\pi}, \quad u_{06} = -\tfrac{1}{16}\sqrt{2\pi}.$$

Both parity blocks of $(W_{mn})_{m,n \le 6}$ are positive definite, with eigenvalues cascading like the zero weights $e^{-\pi\gamma_j/2}$: the matrix is numerically a Gram matrix of the first few zeros, dominated by $\gamma_1 \approx 14.13$ ($W_{00} \approx 2.7 \times 10^{-8}$ is essentially the first zero's weight alone).

4 Remarks

Remark 4.1 (incommensurability). The three constituents live in different worlds: the pole in $\mathbb{Q}\sqrt{\pi}$, the archimedean term in a rank-two lattice over $\mathbb{Q}$, the prime term in the infinite $\mathbb{Q}$-tower spanned by $-\zeta'/\zeta(3), -\zeta'/\zeta(5), \ldots$ PSLQ at 115–240 digits finds no rational relation between the prime entries and the other two sides, nor among the 16 prime entries themselves. The near-vanishing $W_{mn} \approx 0$ at small m, n is therefore an analytic cancellation, not an algebraic identity.

Remark 4.2 (positivity). Under RH every W_{mn} truncation is a Gram matrix, hence positive semidefinite — and by Theorem 1.1 this is a statement about explicit rational-coefficient combinations of $\sqrt{\pi}$, $\sqrt{2\pi}$, $\gamma + \log 8\pi$ and ζ'/ζ at odd integers. The converse (Weil positivity over this family implies RH) would require the span of the h_{mn} to be dense in an admissible class; we do not address this here.

Remark 4.3 (constant of the infinite prime). The single archimedean constant $\gamma + \log 8\pi$ plays the role that $-\zeta'/\zeta(1+2k)$ plays for the finite primes: it is the logarithmic derivative of the archimedean Euler factor averaged against the basis, arising as $\partial_\lambda \log\big[\Gamma(\lambda)\Gamma(\lambda + \frac{1}{2})\pi^{-\lambda}\big]$ at $\lambda = \frac{1}{4}$, up to sign and rationals.